MTH 200 Chapter 4: Applications of Derivatives

4.4 The Mean Value Theorem Worksheet

1. Let $f(x)=2x^2+7$. Determine the value of c which satisfies the conclusion of the *Mean Value Theorem* on the interval $[1,5]$. You must show all work and simplify your answer.

   We know that $f(x)$ is a polynomial and therefore is continuous and differentiable on any interval $[1,5]$.

   Therefore by the *Mean Value Theorem* there is a c in $[1,5]$ such that: $f'(c)=\dfrac{f(5)-f(1)}{5-1}$.

   $$f'(c)=\frac{f(5)-f(1)}{5-1} \quad\rightarrow\quad 4c=\frac{57-9}{4} \quad\rightarrow\quad 4c=12 \quad\rightarrow\quad c=3$$

   The value of c is 3.

2. Prove, using the *Intermediate Value Theorem* and *Rolle's Theorem*, that the function $f(x)=\frac{1}{3}x^3+2x^2+5x-1$ has exactly one real root.

   Show there is a real root
   We know that *f* is a continuous because it is a polynomial. We can see that:
   - $f(0)=-1<0$
   - $f(1)=\dfrac{19}{3}>0$

   Therefore by the IVT there exists a number c in the interval $(0,1)$ such that $f(c)=0$.
   Therefore we have shown that $f(x)$ has at least one real root.

   Show there cannot be two real roots
   Assume that there are two real roots, say $f(c_1)=f(c_2)=0$ (so we are calling c_1 & c_2 roots).
   Since *f* is a polynomial, we know that is continuous on the interval $[c_1,c_2]$ and differentiable on the interval (c_1,c_2), regardless of what the values of c_1 & c_2 are. Therefore, we have satisfied the conditions for Rolle's Theorem so we can then say, there is a number d in (c_1,c_2) such that $f'(d)=0$.
   We note that: $f'(x)=x^2+4x+5$.

   However, when we look to find when $f'(x)=0$ we can see that this is *not* possible, since the discriminate $b^2-4ac<0$. This is a Contradiction!

   Therefore, our initial assumption must be false. There *cannot* be two real roots.
   So we have shown that the function $f(x)=\frac{1}{3}x^3+2x^2+5x-1$ has exactly one real root.

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3. Verify that $f(x)=e^{-2x}$, $[0,3]$ satisfies the hypothesis of the Mean Value Theorem. Then find all *c* that satisfy the conclusion of the Mean Value Theorem.

$f(x)$ is everywhere continuous and everywhere differentiable so it is clearly continuous on $[0,3]$ and differentiable on $(0,3)$. So by the MVT there is a *c* in the interval $(0,3)$ such that $f'(c)=\dfrac{f(3)-f(0)}{3-0}$.

So we want to find all values *c* such that $f'(c)=\dfrac{e^{-6}-1}{3}$. We observe that $f'(x)=-2e^{-2x}$. So when does $-2e^{-2c}=\dfrac{e^{-6}-1}{3}$?

$$-2e^{-2c}=\frac{e^{-6}-1}{3} \rightarrow e^{-2c}=\frac{e^{-6}-1}{-6} \rightarrow -2c=\ln\left(\frac{e^{-6}-1}{-6}\right) \rightarrow c=-\frac{1}{2}\ln\left(\frac{e^{-6}-1}{-6}\right)$$

The value of *c* is $c=-\dfrac{1}{2}\ln\left(\dfrac{e^{-6}-1}{-6}\right)$.

4. Verify that the hypothesis of Rolle's Theorem are satisfied for the function $f(x)=\sqrt{x}-\frac{1}{3}x$ on the interval $[0,9]$, and find all values of *c* in that interval that satisfy the conclusion of the theorem.

Hypotheses

(1) *f* is continuous on $[0,9]$ because *f* is a radical minus a polynomial.

(2) *f* is differentiable on $(0,9)$ because f' only non-differentiable at $x=0$.

(3) $f(0)=0$ & $f(9)=0$

By Rolle's Theorem there is at least one point *c* in $(0,9)$ such that $f'(c)=0$.

$$f'(c)=\frac{1}{2\sqrt{c}}-\frac{1}{3}$$
$$0=\frac{1}{2\sqrt{c}}-\frac{1}{3}$$
$$\frac{1}{3}=\frac{1}{2\sqrt{c}}$$
$$\sqrt{c}=\frac{3}{2}$$
$$c=\frac{9}{4}$$

5. Let $f(x) = \tan(x)$. Show that $f(0) = f(\pi)$ but there is no number c in the interval $(0, \pi)$ such that $f'(c) = 0$. Why does this not contradict Rolle's Theorem?

$f(0) = \tan(0) = 0$
$f(\pi) = \tan(\pi) = 0$

Notice that $f'(x) = \sec^2(x) = \dfrac{1}{\cos^2(x)}$. So $f'(x)$ is undefined when $x = \dfrac{\pi}{2}$.

So *f* is not differentiable on the interval $(0, \pi)$.

6. Use Rolle's Theorem to prove that the function $f(x) = x + \frac{1}{2}\sin(x) - 3$ has exactly one real root.

Show there is a real root

We know that *f* is a continuous because it is a polynomial plus a trigonometric function. We see that:

- $f(0) = -3 < 0$
- $f(\pi) = \pi - 3 > 0$

Therefore by the IVT there exists a number c in the interval $(0, \pi)$ such that $f(c) = 0$.
Therefore we have shown that $f(x)$ has at least one real root.

Show there cannot be two real roots

Assume that there are two real roots, say $f(c_1) = f(c_2) = 0$ (so we are calling c_1 & c_2 roots).
Since *f* is a polynomial plus a trigonometric function, we know that is continuous on the interval $[c_1, c_2]$ and differentiable on the interval (c_1, c_2), regardless of what the values of c_1 & c_2 are. Therefore, we have satisfied the conditions for Rolle's Theorem so we can then say, there is a number *d* in (c_1, c_2) such that $f'(d) = 0$.

We note that: $f'(x) = 1 + \frac{1}{2}\cos(x)$.

However, when we look to find when $f'(x) = 0$ we can see that this is *not* possible.

$$0 = 1 + \frac{1}{2}\cos(x) \quad \rightarrow \quad \cos(x) = -2 \quad \text{Contradiction!}$$

Therefore, our initial assumption must be false. There *cannot* be two real roots.
So we have shown that the function $f(x) = x + \frac{1}{2}\sin(x) - 3$ has exactly one real root.

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7. Use Rolle's Theorem to prove that the function $f(x) = x^4 + 4x + 1$ has at most two real roots.

Show there is a real root
We know that *f* is a continuous because it is a polynomial. We see that:

- $f(0) = 1 > 0$
- $f(-1) = -2 < 0$

Therefore by the IVT there exists a number c in the interval $(-1, 0)$ such that $f(c) = 0$.
Therefore we have shown that $f(x)$ has at least one real root.

Show there cannot be two real roots
Assume that there are three real roots, say $f(c_1) = f(c_2) = f(c_3) = 0$ (so we are calling c_1, c_2, & c_3 roots). Suppose for the sake of argument that $c_1 < c_2 < c_3$. Since *f* is a polynomial, we know that is continuous on the intervals $[c_1, c_2]$ and $[c_2, c_3]$. We also know that *f* is differentiable on the intervals (c_1, c_2) and (c_2, c_3). Therefore, we have satisfied the conditions for Rolle's Theorem (twice) so we can then say, there is a number d_1 in (c_1, c_2) and a number d_2 in (c_2, c_3) such that $f'(d_1) = f'(d_2) = 0$.
We note that: $f'(x) = 4x^3 + 4$.

However, when we look to find when $f'(x) = 0$ we can see that this is only *one* possible value for which $f'(x) = 0$.

$$0 = 4x^3 + 4 \quad \rightarrow \quad x^3 = 1 \quad \rightarrow \quad x = -1 \qquad\qquad \text{Contradiction!}$$

Therefore, our initial assumption must be false. There *cannot* be three real roots.
So we have shown that the function $f(x) = x^4 + 4x + 1$ has at most two real roots.

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8. You are driving on a straight highway with a speed limit is 55 *mi/h*. At 8:05am a police car clocks your velocity at 50 *mi/h* and at 8:10am a second police car posted 5 miles down the road clocks your velocity at 55 *mi/h*. Explain why the police have the right to charge you with a speeding violation?

You traveled 5 miles and the time elapsed in $5\min = \frac{1}{12} hr.$

So, $avg_V = \dfrac{\Delta miles}{\Delta time} = \dfrac{5miles}{\frac{1}{12}hrs} = 60mi/hr$.

Therefore, the Mean Value Theorem guarantees the police that your instantaneous velocity was 60*mph* at least once over the 5 mile section of the highway.